

Addressing Hidden Biases in AI-Assisted Peer Review

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Abstract

Generative AI is increasingly being considered as a means of reducing the growing workload associated with scientific peer review. While AI-assisted reviewing may improve efficiency, it may also introduce biases that are difficult for human reviewers to detect or correct. This article examines three potential sources of bias in AI-assisted peer review: demographic bias, normative bias, and automation bias; these biases are particularly concerning because peer review derives value not from perfect objectivity, but from the independent judgments and diverse perspectives of multiple reviewers. AI assistance may alter not only individual evaluations but also the diversity of perspectives that contribute to scientific decision-making. We argue that responsible adoption of AI-assisted peer review requires evaluating reviewer behavior alongside AI outputs, testing systems for disparate effects across author populations and research practices, and treating AI systems as potentially biased participants in the review process rather than neutral assistants. Preserving the independence and diversity of scientific judgment should be a central consideration as AI becomes more integrated into peer review.

1 Introduction

It is well discussed that scientific publishing is experiencing a growing peer-review burden. In recent years, the number of scientific publications submitted for review has continuously increased, placing increased strain on reviewers during the peer review process. As journals face difficulty recruiting sufficient reviewers, generative artificial intelligence has emerged as a potential means of reducing the labor required to evaluate scientific manuscripts. For example, the ACM Conference on Human Factors in Computing Systems (commonly refereed to as ACM CHI), is piloting AI tools for submission completeness checking, reviewer matching, desk-reject supporting, and assisted desk-reject support¹ during the upcoming review cycle (Cox et al., n.d.).

While AI-assistance in peer review is attractive for reducing the labor intensiveness of the review process, the well documented limitations and concerns, such as hallucinations and biases (Bender et al. 2021; Vasu et al. 2026; Abid et al. 2021) remain a point of concern. The ethical discussion surrounding AI peer review has focused on immediate concerns such as manuscript confidentiality, chances for hallucinated/incorrect results, and the responsibility of human reviewers. These concerns are important, but they may overlook a more fundamental problem. This work argues that the dangers of AI peer reviewers is not that they may produce a bad review, but rather that AI-assisted peer review may change how those judgments are formed: it can introduce or exacerbate biases that are difficult to observe, encourage reviewers to defer to automated recommendations, or favor research that resembles patterns already represented in scientific literature. In doing so, AI-assistance may reduce the independence and diversity of perspectives that make peer review valuable in the first place.

¹ Assisted desk-reject support AI tools will not be used in the upcoming CHI 2027 review cycle. The other tools will be used.

2 Peer Review and AI Assistance

Peer review is not an objective process; it is a human process, where each reviewer brings their own internalized experiences, biases, and pre-conceived ideas into the review process.

Giannakakos et al. (Giannakakos et al. 2025) showed that authors who are male, white, located in the US or North America, well-published or known in their field, and/or affiliated with prestigious institutions tend to have more positive outcomes in the review process.

Nonetheless, the imperfect nature of human judgment does not mean that peer review is without merit. Peer review incorporates safeguards intended to reduce individual biases, including the use of multiple independent reviewers, anonymized reviewing, author rebuttals, and editorial oversight. For example, evidence from a large computer science conference suggests that double-blind review can reduce the influence of author prestige on reviewer evaluations (Sun et al. 2022). These mechanisms do not eliminate bias, but they can provide checks against the influence of any single reviewer’s judgment. Reviewers often disagree (Bornmann 2010), leading to multiple viewpoints informing the revisions made to a manuscript and the final decision of acceptance. Despite its flaws, the peer review process is a long-standing institution that determines which authors get recognized and which works become documented as scientific knowledge.

The introduction of modern AI systems, namely large language models (LLMs), present a new challenge. Large language models are trained on a large corpora of existing text and learn statistical regularities from those data. Thus, when queried about a manuscript, an LLM system may draw upon learned representations of how scholarly research is typically described and evaluated (Vasu et al. 2026; Bornmann 2010). Due to this, it is important to clarify the current state of AI systems in peer review; they, to the best of the author’s knowledge, are not being used as fully autonomous reviewers in reputable publishing sites. Human feedback and judgment seems to remain an important aspect of the peer review process; the aforementioned CHI conference (Cox et al., n.d.) emphasizes the role of humans in verifying AI work and making any final decisions, and tools are not being used as full reviewers — just assistants.

Still, the increasing amount of automation beckons caution; if the number of submitted manuscripts continues to rise, it seems likely that the desire to offload work to AI systems will increase.² This, in turn, may increase the effects of the biases discussed above. Even today, with limited AI involvement and strong human oversight, it stands to reason that some level of bias is being introduced into the research process, made more difficult to quantify due to the "black-box" nature of LLM systems. I argue that this bias is not sufficiently studied given the adoption of AI-assisted peer review.

² It is also worth addressing that individual reviewers may be utilizing LLM systems for their own peer review tasks, whether permitted by conferences/journals or not.

3 Hidden Biases of AI Peer Reviews

3.1 Demographic Bias

Bias based on some demographic marker, such as location, race, or gender, may be one of the most concerning implications for using AI tools in the review process. Vasu et al. (Vasu et al. 2026) experimentally examined LLM-generated peer reviews by manipulating author metadata — such as institutional affiliation, gender, and publication history — and found that institution, seniority, publication record, and gender affected the reviews.

This bias is particularly concerning for the potential effects on already underrepresented populations. If these effects persist in systems used for peer review, researchers from less prestigious institutions or with shorter publication records could face additional disadvantages. It is possible that this could disproportionately harm authors in developing regions, where institutions may not yet have high prestige. Bias against minority religions or racial groups, which researchers have found to be present in LLM systems (Abid et al. 2021), are also unacceptable and may further perpetuate the existing biases in the peer review system (Bornmann 2010).

3.2 Normative Bias

Normative bias describes the tendency to favor work that aligns with the "norm". Models learn representations and associations from large-scale textual corpora, including scholarly discourse. Those representations may encode prevailing linguistic, methodological, and social patterns. When models are used to evaluate research, these learned patterns may influence the criteria or reasoning they generate. However, terms such as “novel,” “rigorous,” and “important” are not objective measurements. Their interpretation depends partly on disciplinary norms and community values, which may shift over time. Still, the values of what valid research is will be entangled within the AI model (Johnson and Verdicchio 2024).

This raises the possibility that unconventional work could be disadvantaged when its contributions do not resemble patterns represented in the literature on which the model learned. However, some of these works may be important to accept for publication for those very reasons — they challenge existing beliefs or introduce new techniques to the field. However, if an AI system evaluates new research primarily according to patterns represented in existing literature, it may be more comfortable with work that resembles what has already been accepted. Whether this effect actually occurs in AI-assisted peer review remains an empirical question.

3.3 Automation Bias

A prevalent issue in human-AI collaboration is automation bias, the likelihood to over-rely on automated recommendations (Romeo and Conti 2025; Skitka et al. 1999). Suppose an AI system provides a reviewer with a preliminary assessment stating that a manuscript lacks novelty or is out of scope for the venue. The reviewer may agree with that assessment after independently reading the manuscript. However, the reviewer may also begin their evaluation from the AI’s conclusion rather than forming an independent judgment. The AI has therefore influenced the decision even though the human reviewer retains final authority. This concern is even more

prevalent when a reviewer faces large workloads; having a cognitive starting point for evaluation becomes a helpful tool, but reduces human judgment in the process.

Human oversight is often presented as a safeguard against AI bias: a reviewer remains responsible for the final evaluation. However, responsibility does not mean control. An AI system can influence which weaknesses a reviewer notices, which interpretations appear plausible, and what standards seem appropriate without formally controlling the final decision.

4 Designing Better AI-Assisted Peer Review

This is not to say that generative AI usage in the research process is impossible to do ethically or without degrading the scientific process. In fact, it is possible that AI could help mitigate the biases of human reviewers. Still, concrete research needs to be done on how AI reviewer systems may affect the reviewing process and norms around its adoption should be challenged. In particular:

- **Studies should evaluate reviewer behavior, not just AI tool outputs.** If reviewers tend to over-rely on AI outputs, it threatens the nature of reviewing as a multi-perspective process and may lead to a homogenization of reviews.
- **Research should evaluate whether AI reviewing tools cause bias towards underrepresented communities and/or unconventional research practices.** If AI-assistance causes disparities to any community, of which underrepresented ones should be particularly noted, then immediate change to AI-assistance is needed.
- **AI review systems should be treated with scrutiny, not as neutral parties.** The ability for AI tools to be biased presents the opportunity for reviewers to unconsciously introduce more bias into their review process. Training data, language used, and failure modes should be well-understood and critiqued by reviewers.

Another possible path forward is pushing for AI to serve as a verification tool after reviewers have conducted their reviews. While this method would not help reduce the workload of the initial review, it could mitigate bias by treating AI tools as an additional check to the current, not the preliminary basis for review.

5 Conclusion

AI-assisted peer review may offer a practical response to the growing demands placed on scientific reviewers. However, AI biases may affect both automated evaluations and the judgments of humans using these systems, raising concerns about their ethical use. Peer review is valuable not because it produces perfectly objective judgments, but because it creates a structured process in which multiple people with different expertise and perspectives evaluate scientific work. Introducing AI into that process changes how judgments are formed.

The central question is whether scientific communities can use AI to reduce the burden of peer review without allowing AI systems to quietly shape what researchers consider valuable, rigorous, and worthy of publication. It is possible that AI assistance may also reduce certain

human biases by providing reviewers with an additional perspective or by drawing attention to methodological issues that reviewers overlook. The relevant question is therefore not whether AI is biased, but whether AI-assisted reviewing produces better and fairer outcomes than human-only review. If AI-assisted peer review becomes widespread, preserving the diversity and independence of scientific judgment should be treated as a key goal, and further work is needed to see how this is possible.

References

- Abid, Abubakar, Maheen Farooqi, and James Zou. 2021. “Persistent Anti-Muslim Bias in Large Language Models.” *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society* (New York, NY, USA), AIES '21, 298–306. <https://doi.org/10.1145/3461702.3462624>.
- Bender, Emily M., Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. “On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?” *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (New York, NY, USA), FAccT '21, 610–23. <https://doi.org/10.1145/3442188.3445922>.
- Bornmann, Rüdiger AND Daniel, Lutz AND Mutz. 2010. “A Reliability-Generalization Study of Journal Peer Reviews: A Multilevel Meta-Analysis of Inter-Rater Reliability and Its Determinants.” *PLOS ONE* 5 (12): 1–10. <https://doi.org/10.1371/journal.pone.0014331>.
- Cox, Anna, Tony Tang, Thomas Kosch, Erin Solovey, Petra Isenberg, and Regan Mandryk. n.d. “AI-Assisted Tools in the CHI 2027 Papers Review Process: What They Do, What They Do Not Do, and How Humans Remain Responsible.” <https://chi2027.acm.org/2026/08/29/ai-assisted-tools-in-the-chi-2027-papers-review-process-what-they-do-what-they-do-not-do-and-how-humans-remain-responsible/>.
- Giannakakos, Vasiliki P., Troy S. Karanfilian, Antonios D. Dimopoulos, and Anne Barmettler. 2025. “Impact of Author Characteristics on Outcomes of Single- Versus Double-Blind Peer Review: A Systematic Review of Comparative Studies in Scientific Abstracts and Publications.” *Scientometrics* 130 (1): 399–421. <https://doi.org/10.1007/S11192-024-05213-X>.
- Johnson, D. G., and M. Verdicchio. 2024. “The Sociotechnical Entanglement of AI and Values.” *AI & Society* 40: 67–76. <https://doi.org/10.1007/s00146-023-01852-5>.
- Romeo, Giuseppe, and Daniela Conti. 2025. “Exploring Automation Bias in Human–AI Collaboration: A Review and Implications for Explainable AI.” *AI & Society* (Berlin, Heidelberg) 41 (1): 259–78. <https://doi.org/10.1007/s00146-025-02422-7>.
- Skitka, Linda J., Kathleen L. Mosier, and Mark Burdick. 1999. “Does Automation Bias Decision-Making?” *International Journal of Human-Computer Studies* 51 (5): 991–1006. <https://doi.org/https://doi.org/10.1006/ijhc.1999.0252>.
- Sun, Mengyi, Jainabou Barry Danfa, and Misha Teplitskiy. 2022. “Does Double-Blind Peer Review Reduce Bias? Evidence from a Top Computer Science Conference.” *Journal of the Association for Information Science and Technology* 73 (6): 811–19. <https://doi.org/https://doi.org/10.1002/asi.24582>.

Vasu, Sai Suresh Macharla, Ivaxi Sheth, Hui-Po Wang, Ruta Binkyte, and Mario Fritz. 2026. “Justice in Judgment: Unveiling (Hidden) Bias in LLM-Assisted Peer Reviews.” In *Findings of the Association for Computational Linguistics: ACL 2026*, edited by Maria Liakata, Viviane P. Moreira, Jiajun Zhang, and David Jurgens. Association for Computational Linguistics.
<https://doi.org/10.18653/v1/2026.findings-acl.14>.